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Chemical Reaction Engineering

CHPE4512

Production of Cumene

Memo 2

Section 02

Group 03

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Chapter 2: Isothermal Reaction Design with Python 3D

2.1 Polymath Isothermal Reactor Model

An isothermal reactor model was developed to estimate the volume of cumene produced, based on the following key assumption:

- 1- No side reaction will occur during the process.
- 2- The pressure drop across the reactor is considered negligible.

The reaction of the Cumene production:

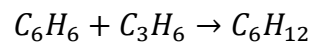


Table 1: Data of the reaction

Typical Temperature	355 °C
Typical Pressure	35 atm
Typical Conversion	99%
Rate Law of Above Reaction	$-r = k C_B^a C_P^b$
Inlet Composition	$y_B = 0.5, y_P = 0.475$
Inert Composition	$y_I = 0.025$
Product rate	100,000 tone/year
Bulk density	1150 Kg/m ³ [1]

2.1.1 Calculation

Calculation at 325 °C

Catalyst weight: 121,300 kg

$$Volume = \frac{Catalyst\ weight}{Bulk\ density}$$

$$Volume = \frac{121,300\ kg}{1,150\ kg/m^3}$$

$$Volume = 105.5\ m^3$$

Calculation at 355 °C

Catalyst weight: 44,610 kg

$$Volume = \frac{Catalyst\ weight}{Bulk\ density}$$

$$Volume = \frac{44,610\ kg}{1,150\ kg/m^3}$$

$$Volume = 38.8\ m^3$$

The volume of the catalyst was calculated based on the mass of the catalyst, as well as a fixed density of 1,150 kg/m³. At a temperature of 325 °C, the required mass of the catalyst is 121,300 kg, which gives a volume of 105.5 m³. Increasing the temperature to 355 °C requires a smaller mass of the catalyst, namely 44,610 kg, which gives a volume of 38.8 m³. This shows that as the temperature increases, the volume of the catalyst decreases. This is because, at higher temperatures, the rates of reaction increase, requiring a smaller amount of the catalyst to produce a given level of production. However, temperatures that are too high need to be chosen carefully, as they might result in an increase of side reactions and costs, as well as the reduction of the life of the catalyst.

The relation between weight and conversion is shown in Figure 1 below.

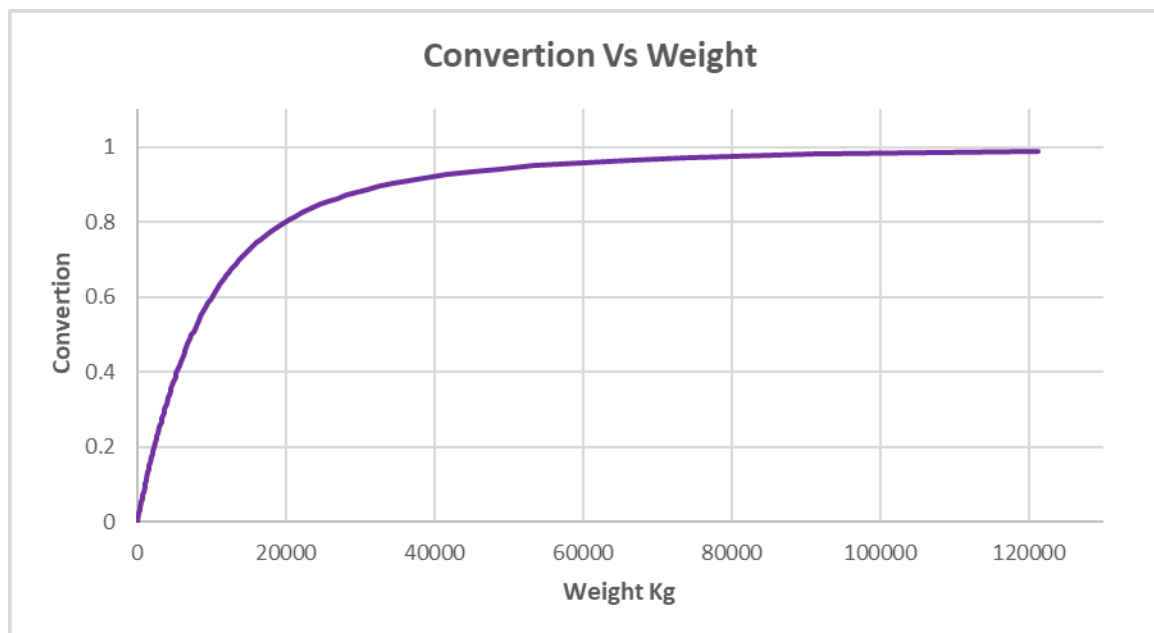


Figure 1: Weight of catalyst vs conversion at T=598.15 k

The figure shows our Previous Temperature 598.15 k relation, Conversion increases steeply for lower weights, while the rate at which the increase slows down reduces with the weight. After 50,000 kg or so, the curve levels off, with the conversion remaining relatively stable at about 1.0. Adding weight beyond this point makes little difference.

2.1.2 Effect of Temperature in Conversion

By altering the temperature in a polymath model, the impact of temperature and catalyst weight on conversion was shown. The figure below was plotted using the conversion and catalyst weight values that were acquired.

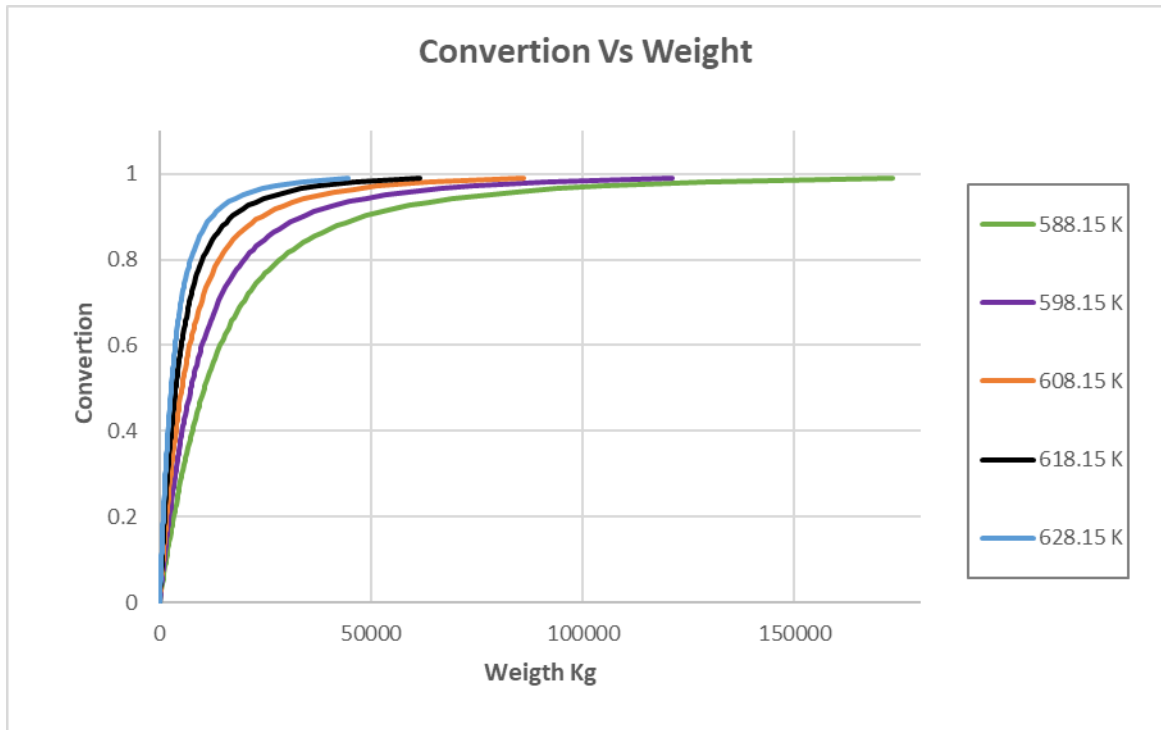


Figure 2: Weight of catalyst vs conversion at different temperatures

The graph shows the performance of our Packed Bed Reactor at different temperatures. As the amount of catalyst increases, the conversion increases because more sites are available to take part in the reaction. When the temperature increases, the reaction occurs faster, meaning that less catalyst weight is required to achieve a given conversion. Hence, we adjusted the temperature from 598 K to 628 K. It was noted that the conversion increased more to 618 K compared to 598 K. This improves the performance of the reactor as high conversion is achieved with less catalyst weight and reactor size. However, high temperatures may mean high energy consumption and may reduce the life of the catalyst.

2.1.3 Applying on Polymath

Polymath code and Table are provided in Appendix B below.

Appendix B:

```
# Mole balance
d(w)/d(x) = (FA0 / rA)

# Rate law
rA = k * cA * cB # Corrected sign

# Stoichiometric relationships
cA = cA0 * ((1 - x) / (1 + eps * x))
cB = cA0 * ((1.05 - x) / (1 + eps * x))

# Parameters
cA0 = 0.33882 # mol/dm^3
R = 8.314 # J/mol.K
FA0 = 27.75 # mol/s
k0 = 2.8 * 10 ^ 7
E = 104174 # J/mol
T = 628.15 # K
k = k0 * exp(-E / (R * T))
yA0 = 0.475
eps = yA0 * (-1)

# Boundary Conditions
x(0) = 0
w(0) = 0
x(f) = 0.99
```

Calculated values of DEQ variables

	Variable	Initial value	Minimal value	Maximal value	Final value
1	x	0	0	0.99	0.99
2	w	0	0	4.461E+04	4.461E+04
3	cA0	0.33882	0.33882	0.33882	0.33882
4	yA0	0.475	0.475	0.475	0.475
5	eps	-0.475	-0.475	-0.475	-0.475
6	cB	0.355761	0.0390479	0.355761	0.0390479
7	R	8.314	8.314	8.314	8.314
8	FA0	27.75	27.75	27.75	27.75
9	k0	2.8E+07	2.8E+07	2.8E+07	2.8E+07
10	E	1.042E+05	1.042E+05	1.042E+05	1.042E+05
11	T	628.15	628.15	628.15	628.15
12	k	0.0608302	0.0608302	0.0608302	0.0608302
13	cA	0.33882	0.0071006	0.33882	0.0071006
14	rA	0.0073324	1.687E-05	0.0073324	1.687E-05

Differential equations

1 $d(w)/d(x) = (FA0 / rA)$

Explicit equations

1 $cA0 = 0.33882$

mol/dm³

2 $yA0 = 0.475$

3 $eps = yA0 * (-1)$

4 $cB = cA0 * ((1.05 - x) / (1 + eps * x))$

5 $R = 8.314$

J/mol.K

6 $FA0 = 27.75$

mol/s

7 $k0 = 2.8 * 10^7$

8 $E = 104174$

J/mol

9 $T = 628.15$

K

10 $k = k0 * \exp(-E / (R * T))$

11 $cA = cA0 * ((1 - x) / (1 + eps * x))$

12 $rA = k * cA * cB$

Corrected sign

2.2 Python 3D Sensitivity Visualization

In the 3D surface plot, it is shown how the conversion varies with temperature and weight of catalyst in the plug flow reactor used in the cumene reaction. Such a plot is useful in finding the best operating conditions in terms of size of reactor and operating temperature. The following fundamental behaviors:

- **Catalyst Site Limitation:** At low catalyst weights, baseline conversion is severely limited by a scarcity of active catalytic sites, regardless of temperature.
- **Kinetic Acceleration:** As the temperature increases, the reaction velocity increases exponentially, creating a steep conversion gradient.
- **Catalyst Loading Effect:** When the weight of the catalyst is increased, more surface area is made available to the reaction, which increases the conversion.

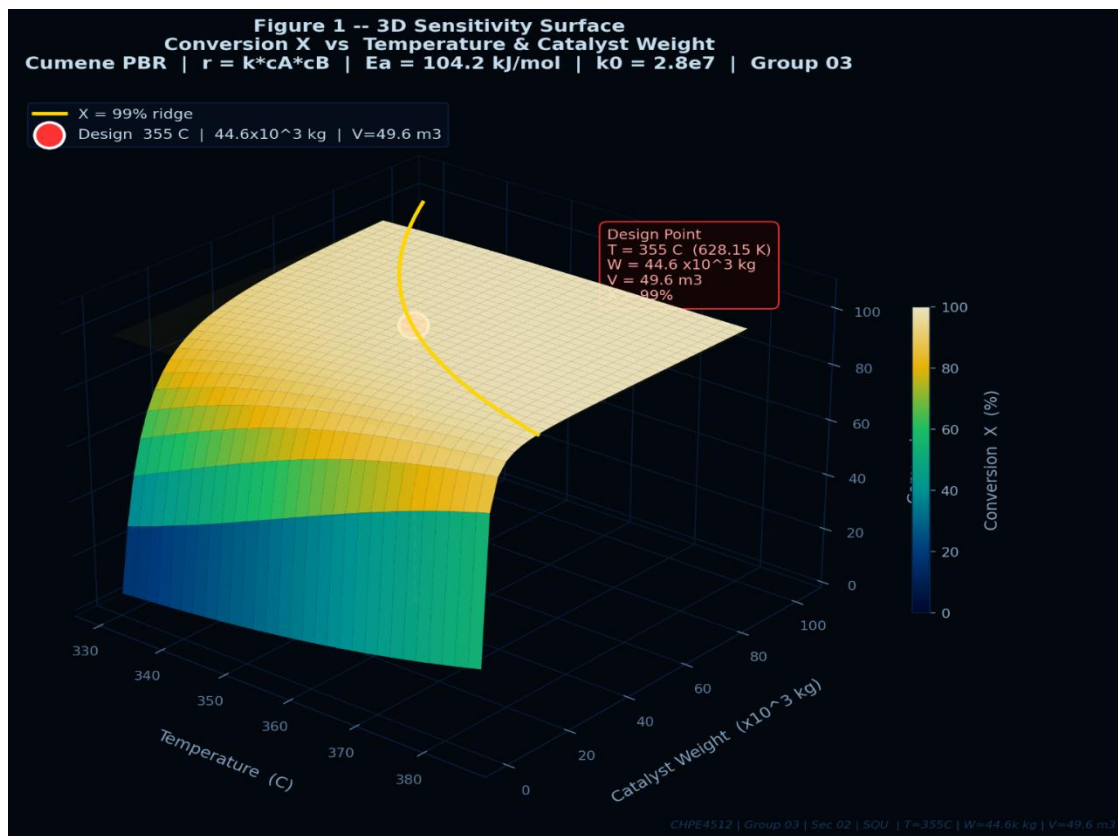


Figure 3: Conversion (X) vs Temperature (T) and Catalyst Weight (W)

2.2.1 Analysis of Conversion Sensitivity to Temperature and Catalyst Weight

Temperature sensitivity

This process is very sensitive to temperature at moderate catalyst weights. Here, an increase in temperature by a small amount greatly accelerates the rate of the reaction due to Arrhenius form of the rate constant. This can be seen from the surface plot where the steepness of the slope is seen along the temperature axis, especially when the weight of the catalyst is not yet sufficient to attain a high level of conversion. With an increase in temperature from about 330°C to 360°C, the level of conversion increases steeply.

Catalyst weight sensitivity

The reaction requires more catalyst weight at lower temperatures. This is because the reaction rate is slower at lower temperatures, and an increase in the weight of the catalyst increases the number of reaction sites. However, when the temperature increases, the effect of the weight of the catalyst diminishes because the reaction rate is high. This means that more catalyst has no effect on the reaction because the reaction is already fast.

The Conversion Plateau

At higher catalyst weights, the surface tends to be relatively flat. This suggests that as the catalyst weight increases, the conversion does not improve significantly. This is because the reaction system is already at a point where the conversion level is approximately at its maximum (99-100%).

- Very high reactant conversion rates that result in low reactant concentrations
- Low reaction rates because of the reaction system being at equilibrium
- Low reaction rates because of the reactants being consumed

Therefore, the reaction system becomes economically less viable as the catalyst weight or the reaction temperature is increased.

2.2.2. Comparison Between Temperature and Catalyst Weight

This is confirmed by the 3D sensitivity surface showing the competitive effect between temperature and catalyst weight in controlling the reactor conversion.

For example, at a temperature of 355 °C or 628 K, an increase in the weight of the catalyst from 35×10^3 kg to 44.6×10^3 kg corresponds to an increase in conversion from 0.95 to 0.99.

This shows that increasing the temperature will significantly minimize the need for the catalyst due to the Arrhenius relationship for the rate constant of the reaction. Consequently, the reactor will have the ability to achieve the desired level of conversion with a smaller volume of the catalyst, thus minimizing the capital cost of the reactor system.

However, decreasing the temperature will require a larger volume of the catalyst to achieve the desired level of conversion. This will result in a larger reactor volume. On the other hand, decreasing the temperature will minimize the energy consumption as well as the stress on the materials of construction of the reactor.

2.2.3 Design Point Selection

The system is getting close to its maximum possible conversion when the chosen design operates in an area where the surface starts to flatten. Reactor performance becomes less sensitive to slight changes in operating conditions, which makes operating in this area beneficial. With a suitable catalyst weight, the reactor achieves almost full conversion under these operating conditions. Further raising the temperature would result in a slight decrease in the need for catalyst, but it would also raise operating expenses and put the reactor and catalyst under more heat stress. In addition, the area around this point on the 3D surface appears relatively flat, suggesting that minor fluctuations in temperature or catalyst weight do not significantly influence the process, thereby showing that the design is stable and suitable for use in industry.

2.2.4 Design Robustness and Potential Improvements

The chosen design is operating in a region where the surface starts to flatten, suggesting that it is close to its maximum achievable conversion. It is advantageous for the system to operate in this region since it is less sensitive to small operating condition changes. If the operation is carried out at a steeper part of the surface, any change in the temperature and catalyst weight can cause significant changes in the conversion, which is not desirable.

A possible improvement could be to increase the temperature slightly, which would result in a smaller weight of the catalyst used and a smaller reactor size. However, this would need to be balanced with the increased power consumption, as well as the deactivation of the catalyst at a

higher temperature. Overall, the selected operating point represents a balanced design, providing high conversion while maintaining reasonable reactor size, operating cost, and process stability.

2.3 Aspen Plus Simulation

In this section, Aspen Plus is used to validate the results obtained previously from the Polymath calculations. The simulation aims to verify whether the reactor volume and catalyst weight computations will result in the same conversion rate as the kinetic study projections. The RPLUG reactor model functions as the selected reactor model for this simulation because it accurately simulates a packed bed reactor which matches our reactor design specifications. The reactor is specified with a catalyst weight of 44,610 kg and a total catalyst volume of 38.8 m³, based on the design calculations at 355 °C. The simulation uses reaction kinetics and rate law which the researchers established in Section 1.2 to maintain identical kinetic parameters for effective validation and comparison.

The design specifications of the reactor include the reactor geometrical dimensions as a necessary component. The reactor diameter measures 2.54 meters while the reactor length extends to 7.62 meters. The actual dimensions of the reactor were increased beyond their original size to include inlet and outlet section measurements which extend from the top to the bottom of the reactor. The simulation evaluates reactor performance after all operating conditions and kinetic expressions and reactor parameters have been defined. The obtained conversion results will be compared to the previous section's conversion calculation for design verification of consistency and accuracy.

2.3.1 Results and Discussion

The reactor simulation for the cumene reaction shows a tubular reactor with 7.8m in length and 2.6m diameter, which means the reactor has a large volume to give the reactants enough space and time to flow and react. The reactor operates at a residence time of 0.613 hour which equals 37 minutes because this time period allows the reactants to remain in the reactor for an intermediate duration, and if this time period becomes shorter the reaction will fail to achieve the necessary conversion but extending the time period will require a larger reactor size which will result in additional side reactions. The reactor operates at constant temperature because its minimum and maximum reactor temperatures both maintain 671°F, which eliminates temperature changes

RPlug	
Name	B1
Process stream property method	PENG-ROB
Process stream Henry's component list ID	
Process stream electrolyte chemistry ID	
Process stream use true species approach for electrolytes	YES
Process stream free-water phase properties method	STEAM-TA
Process stream water solubility method	3
Thermal fluid property method	PENG-ROB
Thermal fluid Henry's component list ID	
Thermal fluid electrolyte chemistry ID	
Thermal fluid use true species approach for electrolytes	YES
Thermal fluid free-water phase properties method	STEAM-TA
Thermal fluid water solubility method	3
Number of tubes	
Reactor dimensions length [meter]	7.8
Reactor dimensions diameter [meter]	2.6
Pressure at reactor inlet: process stream [psi]	0
EO Model components	
Heat duty [Btu/hr]	-597546
Minimum reactor temperature [F]	671
Maximum reactor temperature [F]	671
Residence time [hr]	0.613122
Thermal fluid inlet temperature	
Thermal fluid inlet vapor fraction	
Total feed stream CO2e flow [lb/hr]	0.611839
Total product stream CO2e flow [lb/hr]	0.611839
Net stream CO2e production [lb/hr]	0

Figure 4: Results Summary from ASPEN plus

throughout the reactor. This constant temperature is important because temperature changes directly affect both reaction rate and product quality which results in stable process operation that maintains safety while allowing effective process control through isothermal operation.

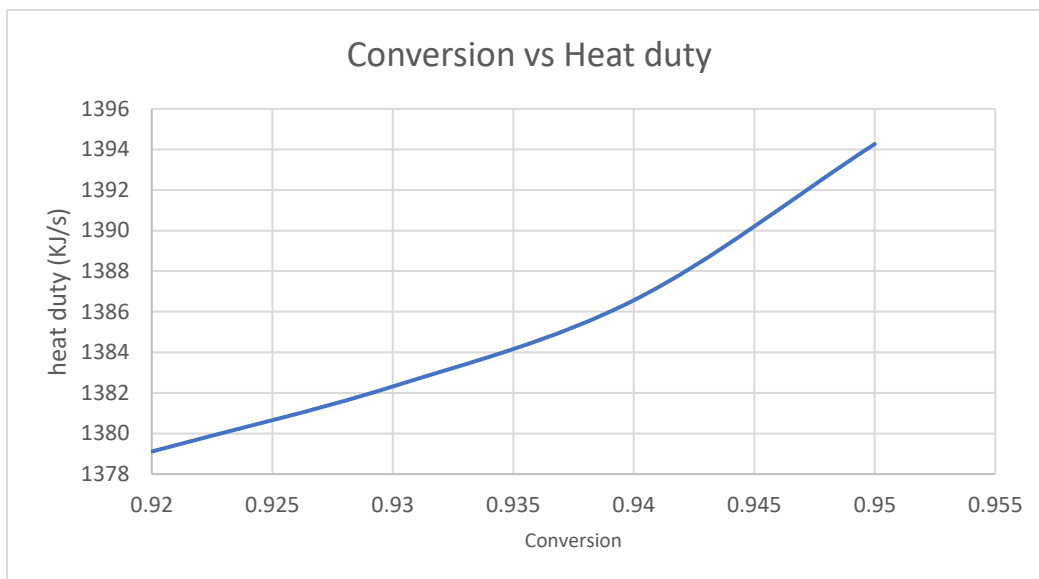


Figure 5: Conversion vs Heat Duty

Based on the kinetic model created with the POLYMATH program, an Aspen Plus simulation with an RPLUG model was created to verify the reactor's thermal performance under an isothermal process at 671F (355C). Figure 5 shows the relationship between the cumulative Propylene conversion and the heat duty required by the reactor. The results obtained from the simulation show that the heat removal increases from 1379 kJ/s to approximately 1394 kJ/s when the conversion increases from 92% to 95%. This is a positive correlation expected for an exothermic reaction to produce Cumene. As the conversion increases, the cumulative enthalpy of the reaction increases, thus requiring higher cooling rates to maintain a constant temperature profile. The upward curvature shows the effect of the cooling utility on the purity of the end product. Operating near the design point of 99% conversion requires a reactor with a high heat management system to maintain stability and avoid catalyst deactivation.

2.3.2 Feed and Product Stream Results

The composition of the reactants changes significantly between the inlet and outlet of the reactor as shown in Table 2, indicating that the alkylation reaction is taking place effectively. The inlet stream contains benzene and propylene, each flowing at a rate of 49.95 kmol/hr and 47.45 kmol/hr, respectively, but no cumene is present in the stream. The outlet stream has no propylene left but contains 47.45 kmol/hr of cumene.

This shows that the conversion of propylene is essentially 100% since there is no propylene left in the outlet stream. The outlet stream is mostly cumene with a mole fraction of 0.90. This shows that the reaction is very selective to the desired product. There is still some benzene that has not been converted. This is expected since benzene is usually added in excess to the reaction to enhance the selectivity of the reaction.

Overall, the results verify that the reactor provides high conversion along with high selectivity to cumene, thus confirming the suitability of the chosen conditions to the reactor design to attain the desired production performance.

Component	Inlet Flowrate (kmol/hr)	Outlet Flowrate (kmol/hr)
Benzene	49.95	2.49639
Propylene	47.4525	0
Cumene	0	47.4536
Propane	2.4975	2.4975
Total	99.9	52.4475

Table 2: Results of Inlet and Outlet from Aspen Plus

References

1. Kordala, N.; Wyszowski, M. Zeolite Properties, Methods of Synthesis, and Selected Applications. *Molecules* 2024, 29, 1069. <https://doi.org/10.3390/molecules29051069>